Practical No.: 1

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Roll No.: 269

Problem Statement: Design a program to maintain a student database that performs the following tasks:

1. Add and store student details (ID, Name, CGPA) using dynamically allocated memory.

2. Expand the student list using realloc() as new entries are added.

3. Implement Linear Search and Binary Search to find student records by ID.

4. Implement at least two Sorting Algorithms (Bubble Sort, Selection Sort, or Insertion Sort) to

sort student records by: a. Name (Alphabetically) b. CGPA (Ascending/Descending)

5.Analyze and compare the performance of search operations before and after sorting.

Code:

```
#include <iostream>
#include <algorithm>
using namespace std;

struct Student
{
   int id;
   string name;
   float cgpa;
};

// Linear Search
int linearSearch(Student s[], int n, int key)
{
   for (int i = 0; i < n; i++)
   {
      if (s[i].id == key)
```

```
        return i;
    }

    return -1;
}

// Binary Search
int binarySearch(Student s[], int n, int key)
{
    int low = 0;
    int high = n - 1;

    while (low <= high)
    {
        int mid = (low + high) / 2;

        if (s[mid].id == key)
            return mid;

        if (key < s[mid].id)
            high = mid - 1;
        else
            low = mid + 1;
    }

    return -1;
}

// Bubble Sort by Name
void bubbleSortName(Student s[], int n)
{
```

```
  for (int i = 0; i < n - 1; i++)
  {
    for (int j = 0; j < n - i - 1; j++)
    {
      if (s[j].name > s[j + 1].name)
        swap(s[j], s[j + 1]);
    }
  }
}

// Selection Sort by CGPA
void selectionSortCGPA(Student s[], int n)
{
  for (int i = 0; i < n - 1; i++)
  {
    int min = i;

    for (int j = i + 1; j < n; j++)
    {
      if (s[j].cgpa < s[min].cgpa)
        min = j;
    }

    swap(s[i], s[min]);
  }
}

int main()
{
  int n;
```

```
cout << "Enter Number of Students : ";
cin >> n;

Student *s = new Student[n];

// Input Student Details
for (int i = 0; i < n; i++)
{
   cout << "\nStudent " << i + 1 << endl;

   cout << "ID : ";
   cin >> s[i].id;

   cout << "Name : ";
   cin >> s[i].name;

   cout << "CGPA : ";
   cin >> s[i].cgpa;
}

// Display Student Records
cout << "\nStudent Records\n";

for (int i = 0; i < n; i++)
{
   cout << s[i].id << " "
        << s[i].name << " "
        << s[i].cgpa << endl;
}

// Linear Search
```

```
int key;

cout << "\nEnter ID to Search : ";
cin >> key;

int pos = linearSearch(s, n, key);

if (pos != -1)
   cout << "Found using Linear Search" << endl;
else
   cout << "Not Found" << endl;

// Sort by Name
bubbleSortName(s, n);

cout << "\nSorted by Name\n";

for (int i = 0; i < n; i++)
{
   cout << s[i].id << " "
        << s[i].name << " "
        << s[i].cgpa << endl;
}

// Sort by CGPA
selectionSortCGPA(s, n);

cout << "\nSorted by CGPA\n";

for (int i = 0; i < n; i++)
{
```

```
        cout << s[i].id << " "
             << s[i].name << " "
             << s[i].cgpa << endl;
    }

    delete[] s;

    return 0;
}
```

Output:

Enter Number of Students : 1

Student 1

ID : 269

Name : Vaishnavi More

CGPA :

Student Records

269 Vaishnavi 0

Enter ID to Search : Not Found

Sorted by Name

269 Vaishnavi 0

Sorted by CGPA

269 Vaishnavi 0